

ASTR-8750: Machine Learning for Physical Sciences

Fall 2025 | from noisy measurements to reproducible scientific claims

Central question: how do we use machine learning to infer, classify, discover, and quantify uncertainty in physical-science data while keeping the science reproducible and interpretable?

Location	Kinard 116 (K-116)	Time	Monday 3:00 - 4:00 PM
Credits	1	Instructor	Srinadh Reddy Bhavanam, Postdoctoral Fellow, Marco Ajello's High-Energy Astrophysics Group
Email	srinadb@clemson.edu	Office hours	K-120A, Friday 10:00 - 11:00 AM

Course Description

This graduate course introduces machine learning as a scientific tool for physics and astronomy. Rather than following a generic survey of algorithms, the course is organized around the workflow that physical scientists actually use frame a scientific question, understand the data and measurement process, build strong baselines, learn from tabular/image/spectral/time-series data, quantify uncertainty, and turn results into a reproducible scientific claim. Examples will be drawn primarily from astrophysics, with transfer to neighboring data-rich physical sciences.

Course Arc

1	2	3	4	5	6
Scientific question	Data and measurement	Baselines and validation	Representation learning	Uncertainty and calibration	Reproducible claim
Prerequisites: Working knowledge of linear algebra, calculus, and probability/statistics. Basic Python and Jupyter familiarity is recommended; a short Python bootcamp will be provided during the first part of the semester.			Software stack: Python, Jupyter, NumPy, pandas, matplotlib, scikit-learn, PyTorch, MLflow, and Git/GitHub.		

Learning Goals

- Translate physical-science questions into regression, classification, clustering, anomaly-detection, or parameter-estimation tasks
- Build and compare baseline and advanced ML models using scientifically honest train/validation/test protocols
- Evaluate models with domain-aware metrics, calibration checks, and error analysis rather than accuracy alone
- Use uncertainty-aware tools including Bayesian neural networks, ensembles, and conformal prediction to communicate confidence responsibly
- Produce reproducible, auditable workflows with clear notebooks, tracked experiments, and publication-ready figures

Distinctive Emphasis

- **Workflow-first structure.** The course follows the path from scientific question to defensible claim, not a generic model-zoo sequence.
- **Uncertainty quantification.** Special attention to calibration, Bayesian neural networks, and conformal prediction for trustworthy ML.
- **Reproducibility.** Experiment tracking with MLflow and clear notebook practices are treated as core research skills, not optional extras.

Topical Flow

The semester moves from scientific framing to uncertainty-aware, reproducible research practice. This flow is intentionally designed for machine learning in the physical sciences rather than as a generic algorithm survey.

Time	Theme	Focus and tools
Weeks 1-2	Scientific ML workflow and data realism	Problem framing; observational vs simulated data; labels, leakage, train/validation/test splits; measurement error; baseline sanity checks; Python refresher.
Weeks 3-4	Feature representations and preprocessing	Tabular features, spectra, images, and time series; scaling/normalization; missing data; class imbalance; feature engineering; simple scientific baselines.
Weeks 5-6	Supervised learning for inference	Linear/logistic regression, trees, random forests, support vector machines; regression vs classification; residual analysis; confusion matrices; ROC-AUC; RMSE/MAE.
Weeks 7-8	Unsupervised discovery and rare-event search	PCA, clustering, density-based methods, anomaly detection, latent structure, and exploratory workflows for survey and simulation data.
Weeks 9-10	Neural networks for scientific data	Perceptrons and MLPs; optimization basics; regularization; monitoring; overview of architectures for images, spectra, and time-series data.
Weeks 11-12	Uncertainty quantification and calibrated prediction	Aleatoric vs epistemic uncertainty; calibration; ensembles; Bayesian neural networks; conformal prediction intervals/sets; uncertainty-aware reporting.
Week 13	Reproducibility and experiment tracking	MLflow for tracking runs, metrics, parameters, and artifacts; model comparison; keeping notebooks, figures, and decisions auditable.
Weeks 14-15	Project studio and presentation	Project proposal refinement, milestone feedback, interpretation of results, communication of failure modes, final presentation, and scientific storytelling.

Assessment and Final Project

Assessment	Final Project Deliverables
• Coding notebooks / mini-labs (25%). Short exercises on Python, arrays, data handling, linear algebra, statistics, and model evaluation. • Project proposal + milestone (15%). Scientific question, dataset choice, baseline plan, and a mid-semester progress update. • Final project (60%). End-to-end physical-science ML analysis with reproducible code, uncertainty discussion, and scientific interpretation.	• Reproducible notebook. Data loading, preprocessing, model training, validation, figures, and clearly documented assumptions. • Tracked experiments. MLflow log (or equivalent run record) for parameters, metrics, artifacts, and model comparisons. • Report + presentation. 5-8 page paper-style report and a short in-class presentation emphasizing results, uncertainty, and scientific meaning.

Milestone dates will be announced in class. Representative projects may include classification of astrophysical sources, regression on physical parameters, anomaly detection in survey data, or surrogate models for simulations.